18. Single-World Intervention Graphs (SWIGs)

A consolidated, mathematical lecture companion

English notes based on the lecture slides *Single World Intervention Graphs: Unifying the Counterfactual and Graphical Approaches to Causality* (T. Richardson, joint work with J. Robins; used in the GSDS SNU course of Prof. Sanghack Lee)

The central question is how potential outcomes can be placed on a causal graph, so that counterfactual independences can be read off by d-separation, without the extra untestable assumptions of a fully specified structural-equation model.

These notes consolidate the slides, define node splitting and the SWIG construction, state factorization, modularity and d-separation on SWIGs, resolve the "harder inferential problem", and prove the adjustment formula. Physical PDF page numbers are listed in Section 7.

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1 Potential outcomes and graphs

Purpose. Recall the two languages that SWIGs unify.

1.1 Potential outcomes

For binary treatment X and response Y: $Y(x=0)$ (also $Y(x_0)$) is the value of Y that would be observed for a unit assigned $X=0$; $Y(x=1)$ (or $Y(x_1)$) likewise for $X=1$. This presumes the outcomes are well defined: one version of treatment, and no interference between units (SUTVA). "Potential outcome" and "counterfactual" are used synonymously.

Response types for binary Y:

$Y(x_0)$	$Y(x_1)$	Name
0	0	never recover
0	1	helped
1	0	hurt
1	1	always recover

Relating counterfactuals and do-notation. $P(Y(x)=y) \equiv P(Y=y \mid \mathrm{do}(X=x))$, but counterfactual notation is more general. Example: the distribution of outcomes that would have arisen among those who took treatment ($X=1$) had they, counter to fact, not received it: $P(Y(x=0)=y \mid X=1)$. If treatment is randomized, $X \perp\!\!\!\perp Y(x=0)$ and this equals $P(Y(x=0)=y)$; in an observational study they may differ.

1.2 Graphs

A DAG is associated with the factorization $P(\mathbf{V}) = \prod_{X \in \mathbf{V}} P(X \mid \mathrm{pa}(X))$; e.g. for $A \to B$, $A \to C$, $B \to D$, $C \to D$, $D \to E$, $P(A,B,C,D,E) = P(A)P(B \mid A)P(C \mid A)P(D \mid B,C)P(E \mid D)$. d-separation reads independences such as $C \perp\!\!\!\perp B \mid A$, $D \perp\!\!\!\perp A \mid B,C$, $E \perp\!\!\!\perp A,B,C \mid D$.

In the graphical approach, edges represent direct causal relations, and by convention confounders (e.g. an unobserved H with $X \leftarrow H \to Y$) are included. It originates in Sewall Wright's path diagrams of the 1920s. For $X \to Y$ without confounding, $P(x,y) = P(x)P(y \mid x)$ and the causal model asserts

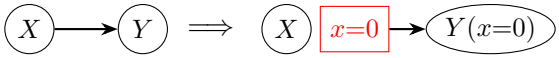
$$P(Y(x)=y) = P(Y=y \mid \mathrm{do}(X=x)) = P(Y=y \mid X=x),$$

where $P(y \mid \mathrm{do}(x))$ is the distribution under an experiment fixing X to x. How does this relate to the counterfactual approach? No confounding corresponds to $X \perp\!\!\!\perp Y(x_0)$ and $X \perp\!\!\!\perp Y(x_1)$; confounding to $X \not\perp\!\!\!\perp Y(x_0)$ or $X \not\perp\!\!\!\perp Y(x_1)$. The elephant in the room: $Y(x_0)$ and $Y(x_1)$ do not appear on these graphs.

2 Node splitting

Purpose. Put counterfactual variables on the graph.

Setting X to 0. Start from $P(X=\tilde{x}, Y=\tilde{y}) = P(X=\tilde{x})P(Y=\tilde{y} \mid X=\tilde{x})$ for $X \to Y$. Split X into a random node X and a fixed node $x=0$; the outcome becomes $Y(x=0)$:



Now $X \perp\!\!\!\perp Y(x=0)$ can be read off, and a new factorization is associated:

$$P(X=\tilde{x}, Y(x=0)=\tilde{y}) = P(X=\tilde{x})\,P(Y(x=0)=\tilde{y}), \qquad P(Y(x=0)=\tilde{y}) = P(Y=\tilde{y} \mid X=0).$$

The last equation links a term of the original factorization to the new one: the *modularity assumption*.

Marginals represented by SWIGs are identified. The SWIG $\mathcal{G}(x_0)$ represents $P(X, Y(x_0))$ and $\mathcal{G}(x_1)$ represents $P(X, Y(x_1))$; under no confounding both are identified from $P(X, Y)$. In contrast $P(X, Y(x_0), Y(x_1))$ is not identified. $Y(x=0)$ and $Y(x=1)$ never appear on the same graph: although $X \perp\!\!\!\perp Y(x=0)$ and $X \perp\!\!\!\perp Y(x=1)$, we do not assume $X \perp\!\!\!\perp \{Y(x=0), Y(x=1)\}$. A single graph containing both would have been impossible; hence *single-world* intervention graphs.

Templates. Both graphs are represented by a template $\mathcal{G}(x)$: $X \mid x \to Y(x)$. Formally the template is a graph-valued function (not a graph): it takes a value x and returns a SWIG $\mathcal{G}(x)$; each instantiation represents a different margin.

Intuition (Robins, VanderWeele and Richardson 2007). How could we learn whether someone would choose treatment ($X=1$) and, at the same time, what happens to that person without treatment ($Y(x=0)$)? Imagine that whenever a patient is observed to swallow the drug ($X=1$), we instantly administer a safe emetic that makes them regurgitate the pill before any drug enters the bloodstream. With no side effects of the emetic, the recorded outcome is $Y(x=0)$, observed jointly with the natural choice X.

3 SWIG construction and properties

Purpose. Construct SWIGs in general and state what they imply.

Given $\mathcal{G}$ and a set $\mathbf{A} = \{A_1, \ldots, A_k\}$ to intervene on, form $\mathcal{G}(\mathbf{a})$ in two steps:

1. **Node splitting:** split every $A \in \mathbf{A}$ into a random node A and a fixed node a. The random half inherits all edges directed *into* A; the fixed half inherits all edges directed *out of* A.
2. **Relabelling:** every descendant Y of a fixed node is relabelled $Y(\mathbf{a}_Y)$, indexed by the fixed nodes that are its ancestors (e.g. $B(a,\ldots)$, $C(a,\ldots)$, $D(a,\ldots)$); non-descendants keep their names.

Factorization and modularity. The original graph $\mathcal{G}$ carries the observed $P(\mathbf{V})$; the SWIG $\mathcal{G}(\tilde{\mathbf{a}})$ carries the counterfactual distribution $P(\mathbf{V}(\tilde{\mathbf{a}}))$.

- *Factorization:* $P(\mathbf{V}(\tilde{\mathbf{a}}))$ factorizes with respect to $\mathcal{G}(\tilde{\mathbf{a}})$, ignoring fixed nodes.
- *Modularity:* the conditional density of $Y(\tilde{\mathbf{a}}_Y)$ in $\mathcal{G}(\tilde{\mathbf{a}})$ equals that of Y in $\mathcal{G}$ after substituting $\tilde{a}_i$ for any $A_i \in \mathbf{A}$ that is a parent of Y.

Consequence: if $P(\mathbf{V})$ is observed, $P(\mathbf{V}(\tilde{\mathbf{a}}))$ is identified.

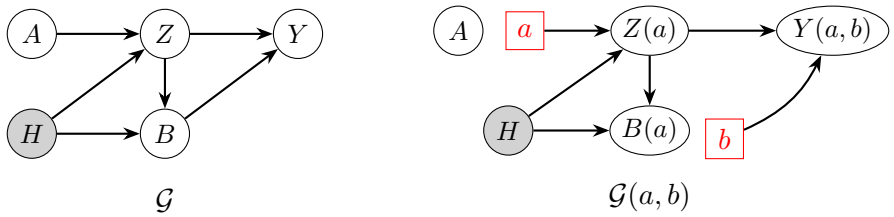
d-separation on SWIGs. If subsets $\mathbf{B}(\tilde{\mathbf{a}})$ and $\mathbf{C}(\tilde{\mathbf{a}})$ of random nodes are d-separated by $\mathbf{D}(\tilde{\mathbf{a}})$ together with the fixed nodes $\tilde{\mathbf{a}}$ in $\mathcal{G}(\tilde{\mathbf{a}})$, then

$$\mathbf{B}(\tilde{\mathbf{a}}) \perp\!\!\!\perp \mathbf{C}(\tilde{\mathbf{a}}) \mid \mathbf{D}(\tilde{\mathbf{a}}) \quad \text{in } P(\mathbf{V}(\tilde{\mathbf{a}})).$$

4 A harder inferential problem

Purpose. Use a SWIG to settle a counterfactual independence query.

Graph $\mathcal{G}$: $A \to Z$, $Z \to Y$, $Z \to B$, $B \to Y$, and an unobserved H with $H \to Z$, $H \to B$. Query: does $\mathcal{G}$ imply $Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A$?



$\mathcal{G}$ $\mathcal{G}(a,b)$

Pearl (2009, Ex. 11.3.3) claims the DAG does not imply

$$Y(a,b) \perp\!\!\!\perp B \mid Z, A = a. \tag{3}$$

The SWIG shows that (3) does hold. In $\mathcal{G}(a,b)$, every path from $B(a)$ to $Y(a,b)$ passes through $Z(a)$ as a non-collider ($B(a) \leftarrow Z(a) \to Y(a,b)$, $B(a) \leftarrow H \to Z(a) \to Y(a,b)$), and $B(a)$ has no edge to $Y(a,b)$ because the edge $B \to Y$ now leaves the fixed node b. Hence

$$Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A \quad \Longrightarrow \quad Y(a,b) \perp\!\!\!\perp B(a) \mid Z(a), A = a,$$

which is equivalent to (3) by consistency ($B(a) = B$ and $Z(a) = Z$ when $A = a$). (Pearl had inferred that a claim of Robins was false because, if true, (3) would hold.)

5 Adjustment for confounding

Purpose. Prove the back-door formula in the SWIG language.

Graph: $L \to X$, $L \to Y$, $X \to Y$. In the SWIG $\mathcal{G}(x)$: $L \to X$, $L \to Y(x)$, $x \to Y(x)$, so $X \perp\!\!\!\perp Y(x) \mid L$. Then

$$\begin{aligned} P[Y(\tilde{x}) = y] &= \sum_l P[Y(\tilde{x}) = y \mid L = l] P(L = l) \\ &= \sum_l P[Y(\tilde{x}) = y \mid L = l, X = \tilde{x}] P(L = l) && (X \perp\!\!\!\perp Y(\tilde{x}) \mid L) \\ &= \sum_l P[Y = y \mid L = l, X = \tilde{x}] P(L = l) && \text{(consistency)}. \end{aligned}$$

Connection to do-calculus. Factorization and modularity suffice to imply all identification results of Pearl's (1995) do-calculus (see also Spirtes et al. 1993): $P(Y = y \mid \text{do}(A = a))$ is identified iff $P(Y(a) = y)$ is identified.

6 Conclusion

Eliminating a false trichotomy. Previously the only unification of counterfactuals and graphs was Pearl's non-parametric structural equation models with independent errors, which left three options: use graphs without counterfactuals (Dawid); use counterfactuals without graphs (many statisticians); or use both but accept many additional assumptions that are not experimentally testable even in principle and not needed for most identification results. SWIGs allow graphs and counterfactuals together without these extra assumptions.

Summary and extensions. SWIGs unify graphs and counterfactuals via node splitting, working by linking the factorizations of the two graphs. The new graph represents a counterfactual

distribution identified from the distribution of the original DAG, giving a common language for counterfactual and graphical researchers. (Not covered: combining information on the absence of individual- and population-level direct effects; models in which interventions on only some variables are well defined.)

References: Pearl (2009); Richardson and Robins (2013), CSSS Technical Report 128; Robins (1986); Robins, VanderWeele and Richardson (2007); Shpitser and Pearl (2008); Spirtes, Glymour and Scheines (1993).

7 Source coverage

The source deck has 26 physical pages.

Topic	Physical pages
Potential outcomes, response types, do-notation	1–5
DAG factorization, graphical approach, linking the two	6–9
Node splitting, identified marginals, templates, emetic intuition	10–13
Harder problem, construction, factorization, d-separation, resolution	14–19
Adjustment, do-calculus connection, conclusion, references	20–26